



BY AMETECs

SR-06

Licence Change Flows — Session Report

Upgrades, Downgrade-at-Renewal, Perpetual Upgrades & Self-Service .lic · 12
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Contents

- 01 Executive Summary**
- 02 What Ejaz Reported vs. What The Code Actually Had**
- 03 The Policies (Decisions D1–D4)**
- 04 What Was Built — Admin Billed Upgrade**
- 05 What Was Built — Downgrade-at-Renewal**
- 06 What Was Built — Perpetual Upgrade**
- 07 What Was Built — Self-Service .lic Re-Download**
- 08 Files Changed (Central face8e7)**
- 09 Automated Test Coverage**
- 10 What YOU Need To Do Now — Exact Instructions**
- 11 Deferred — Agreed Follow-Ups**
- 12 Risks & Notes**

1. Executive Summary

Ejaz reported that the commercial lifecycle of a licence **after purchase** was incomplete: "tenant upgrade and degrade or increase seats — how, and how to bill that — not found in Central nor in product, nor in the client login... and how to re-issue an updated .lic." The investigation confirmed a partial truth: one flow existed but was hidden, and three were genuinely missing. This session closed all of them in one Central commit:

Repo	Commit	Content
Central (smartept-central)	<code>face8e7</code>	Licence change flows — 9 files, +524 lines: admin billed upgrade, downgrade-at-renewal, perpetual upgrade, self-service .lic

The four policies locked with Ejaz before building (formal decisions, in-session): perpetual licences are **upgrades only — never reduced, never refunded** ("once purchased"); a perpetual upgrade is priced as the **difference of interpolated lifetime prices** (the 11-Aug progressive engine) with AMC re-basing automatically; cloud seat **reductions happen only at renewal**, schedulable by both the client and the admin, never below devices in use; and all of it was built together in this session.

Status: committed locally, **not yet migrated, tested or deployed** — Ejaz's exact steps are in Section 10. Every flow is locally checkable in Laragon first.

2. What Ejaz Reported vs. What The Code Actually Had

The survey (before any code was written) found:

Capability	Status before this session
Cloud subscription: add seats mid-period	EXISTED but hidden — built 6-Aug ("Phase 5"): a client-portal card visible only when the licence is <code>subscription + active + days remaining</code> . Trial, expired and perpetual clients saw nothing, and there was no admin-side equivalent
Admin raises a billed upgrade for a client	MISSING — the admin could only <i>silently edit</i> the device limit (no order, no GST invoice, no paper trail)
Downgrade / reduce seats	MISSING entirely — the code said "reductions apply at renewal" but renewal always billed the current size; no field, no UI, no mechanism anywhere
Perpetual (lifetime) licence: change after purchase	HARD-BLOCKED — <code>createUpgradeOrder</code> threw "Only an active Cloud subscription can be upgraded"; no pricing rule, no order type
Re-issue an updated .lic	ADMIN-ONLY — Central admin's "Licence file" button re-signs with the remembered fingerprint (built 6-Aug), but it was disconnected from billing and the client could not self-download

So the honest answer to "why didn't I find it": the one existing flow lived only behind a conditional card in the client portal, and everything else did not exist.

3. The Policies (Decisions D1–D4)

#	Decision	Ejaz's ruling
D1	Perpetual licences after purchase	Upgrades only. No downgrade, no refund, ever. ("once purchased")
D2	Perpetual upgrade price	Difference of interpolated lifetime prices: <code>price(new users) - price(current users)</code> from the progressive-pricing engine approved 11-Aug, +GST. AMC re-bases automatically — the next AMC bill is 18% of the prevailing price at the licence's user count at that time
D3	Cloud downgrades	At renewal only , scheduled by client self-service <i>or</i> admin; never below devices currently in use; mid-period reductions stay impossible (the current period is already paid)
D4	Build scope	All three phases in this session

Also ruled in-session: **plan changes** are meaningless until a second plan exists (SmartEPT currently has one all-features plan) — build the switch flow when a second plan/tier is introduced; **storage top-up self-service** is agreed as the next piece (engine exists; portal buy button missing); **pause subscription** waits for a policy decision (see Section 11).

4. What Was Built — Admin Billed Upgrade

New endpoint `POST admin/api/licences/{id}/upgrade-order` (super + sales) and an **"Upgrade..."** button on every active licence row in Central → Licences. It uses the *same* pricing engines the client portal uses, so an admin-raised upgrade carries the full order + GST paper trail instead of a silent limit edit:

- **Cloud subscription** → pro-rata: `(new × rate_new - old × rate_old) × remaining months`, expiry unchanged, renewals bill the new size.
- **Perpetual** → one-time lifetime price difference (Section 6).

The modal explains which rule applies, prices the order, and hands back the order number, total (incl. GST) and a copy-able pay link; a client NEFT/UPI payment is recorded from Billing as usual. Seats rise the moment the order is paid — through the same `provisionIfNeeded` golden path as every other payment. The action is audit-logged (`licence.upgrade_order`) and appears in the licence History timeline.

5. What Was Built — Downgrade-at-Renewal

A new nullable column `licences.renewal_device_limit` (migration `2026_08_12_000500`) holds the **scheduled reduction**. Three surfaces set it:

1. **Client portal** → Buy & Renew → "Reduce users at next renewal" card: schedule or cancel; input is bounded to at least the devices currently in use and below the current limit. Audited as `client.schedule_reduction`.
2. **Admin** → Edit licence → "Reduce to (at next renewal)" field (blank clears), with the same not-below-active-devices guard.
3. Both show the pending state: the admin licence row shows "↓ **N at renewal**"; the portal licence card shows "reduces to N at renewal".

Mechanics: `createRenewalOrder` bills `renewal_device_limit ? : device_limit`, notes the reduction in the order description, and carries `apply_device_limit` in the order meta. On payment, provisioning renews the licence, **applies the reduced limit, and clears the schedule** — a consumed or abandoned schedule can never haunt a later renewal. The legacy grandfather price-cap logic is unaffected.

6. What Was Built — Perpetual Upgrade

`BillingService::createPerpetualUpgradeOrder(licence, newLimit)`:

- Guards: perpetual + active only; `newLimit` must exceed the current limit ("a lifetime licence never reduces once purchased"); both the current and the new count must be automatically priceable — a count above the last priced milestone, or a legacy count the bands can't price, gets a clear "custom quotation — WhatsApp us" error instead of a wrong number. Never ₹0.
- Price: `calculateLifetimeLicencePrice(plan, new)['price'] - calculateLifetimeLicencePrice(plan, current)['price']` — the same interpolated engine as /buy, the portal calculator and the landing widget, so every surface quotes the same rupee. Worked example on Ejaz's bands (15 → ₹15,000 · 30 → ₹25,000): upgrading 15 → 30 users = **₹10,000 + ₹1,800 GST = ₹11,800**.
- The order uses line type `upgrade` (same GST invoice series), currency conversion for USD tenants, and the standard pay/quote machinery. On payment the existing provisioning branch raises `device_limit` immediately.
- **AMC re-base is automatic by design:** `amcQuote` prices AMC from the licence's user count at billing time, so the next AMC cycle simply bills 18% of the new interpolated price — nothing to remember.

Client-side, the portal's upgrade endpoint now branches by kind, and perpetual tenants get their own card: "Add users to your lifetime licence" with the once-purchased-never-reduced note.

7. What Was Built — Self-Service .lic Re-Download

New endpoint `POST client/api/licences/{id}/license-file` and a **"Download licence file (.lic)"** button on the portal Licences page (client-hosted, active licences). It re-signs a fresh licence file — carrying the *current* entitlements, so it's the natural last step after a perpetual upgrade — with one deliberate security property:

It only ever signs with the fingerprint Central already has stored. A client can re-download for their known machine any time, but can never point the licence at a *new* machine — machine moves remain the admin's investigated "Shift machine" action (the 6-Aug control that checks recent check-ins stays intact). A licence with no stored fingerprint gets a clear message (validate online once, or WhatsApp the fingerprint). Downloads are audit-logged (`licence.file_issued` , by `client:<email>`) and appear in the licence History.

The full perpetual-upgrade journey is now: portal → Add users → pay the difference → seats rise at Central instantly → Licences → Download .lic → import on the server's Licence screen. Zero manual steps at Ametecs.

8. Files Changed (Central `face8e7`)

File	Change
<code>database/migrations/2026_08_12_000500_...</code>	NEW — <code>licences.renewal_device_limit</code> (nullable unsigned int; no ENUMs)
<code>app/Models/Licence.php</code>	<code>renewal_device_limit</code> fillable
<code>app/Services/BillingService.php</code>	Renewal bills scheduled reduction + <code>apply_device_limit</code> meta; provisioning applies + clears the schedule; NEW <code>createPerpetualUpgradeOrder</code>
<code>app/Http/Controllers/Admin/LicenceApiController.php</code>	NEW <code>upgradeOrder</code> (billed, kind-aware); <code>update()</code> accepts/guards <code>renewal_device_limit</code>
<code>app/Http/Controllers/Client/PortalApiController.php</code>	<code>upgrade</code> branches by kind; NEW <code>scheduleReduction</code> ; NEW <code>licenseFile</code> (stored-fingerprint only); payloads expose <code>renewal_device_limit</code> , <code>has_fingerprint</code> , <code>devices_active</code>
<code>routes/web.php</code>	3 new routes (2 client, 1 admin)

File	Change
resources/views/admin/console.blade.php	"Upgrade..." button + modal; "↓ N at renewal" badge; Edit-licence reduction field
resources/views/client/portal.blade.php	Perpetual upgrade card; reduce-at-renewal card; why-can't-I-upgrade note; .lic download button + JS
tests/Feature/LicenceChangeFlowsTest.php	NEW — 5 tests (Section 9)

Verified in-session: `php -l` clean on every PHP file; `node --check` clean on both consoles' inline JS (one real bug caught and fixed: an over-escaped quote in the portal template). **Not run:** migration + test suites — no PHP available through the device bridge; that is Ejaz's Step 1.

9. Automated Test Coverage

tests/Feature/LicenceChangeFlowsTest.php — five tests on Ejaz's example band configuration:

1. **Interpolated difference maths:** 15 → 30 users prices at exactly ₹10,000 subtotal + ₹1,800 GST, and paying the order through `markPaid` (the golden path) raises the limit to 30.
2. **Reduction rejected** on perpetual (once purchased, never reduced).
3. **Above-priced-bands rejected** → custom-quote error, never a wrong price.
4. **Scheduled reduction end-to-end:** renewal order billed at 25 (not 40), description carries the reduction note, payment applies 25 and clears the schedule.
5. **Plain renewal regression:** no schedule → identical behaviour to before.

Manual coverage: **SR-07** (Excel tracker + PDF), 39 cases across 8 areas (21 gating P1s), each with local-Laragon-runnable steps.

10. What YOU Need To Do Now — Exact Instructions

Step 1 — Local verify (Laragon, `C:\laragon\www\smartept-central`):

```
php artisan migrate
php artisan test --filter=LicenceChangeFlowsTest
php artisan test --filter=PerpetualProgressivePricingTest
```

Then Laragon **Stop All** → **Start All** (OPcache).

Step 2 — Local click-through (the ordered list is also in SR-07 as executable cases): admin Upgrade... on a subscription licence → record payment → limit rises; same on a perpetual licence → 15 → 30 must show ₹11,800 total; Edit licence reduction field → "↓ N at renewal" badge; portal

cards (perpetual upgrade / reduce-at-renewal / why-not note); portal .lic download with and without a stored fingerprint; full downgrade journey: schedule → renew → record payment → limit drops + schedule clears.

Step 3 — Push Central `face8e7` (sits on top of `76ddf09` from earlier tonight).

Step 4 — Live deploy (Central on the VPS):

```
git pull
php artisan migrate --force
php artisan view:clear && php artisan config:clear
```

Step 5 — Live smoke: admin Upgrade... prices correctly against the LIVE bands; Khan's portal shows the new cards; a .lic re-download works for a bound client-hosted licence. Then execute SR-07; any P1 fail comes to me immediately.

Reminder — earlier tonight's work is a separate deploy: the per-tenant licensing build (Product `1fcb7f7` + Central `76ddf09`, documented in SR-04/SR-05) still needs its own local test + migrate + deploy on the **product** repo. Central deploys once and covers both `76ddf09` and `face8e7`.

11. Deferred — Agreed Follow-Ups

1. **Storage top-up self-service** (next session): the pricing engine (storage slabs, monthly rate) and the admin record-storage action exist; what's missing is the client-portal buy button + order line type. Agreed to build next so tonight's commit stays reviewable.
2. **Pause subscription:** needs Ejaz's policy first — during a pause, do agents stop? is data retained and for how long? does the expiry date freeze? any credit? I will draft the policy options as a short decision note before writing code.
3. **Plan changes:** build when a second plan/tier exists.
4. Noted while surveying (not changed): AMC renewal has no client-side billed path (the portal says "through our team"), and the admin "Renew AMC" action extends the date without an order. Worth a billed-AMC flow when convenient.

12. Risks & Notes

1. The perpetual upgrade prices from the **admin-saved bands at upgrade time** — if band prices have risen since the client's original purchase, the difference reflects today's curve (that is the agreed rule: "prevailing price"). Grandfathered/legacy counts the bands cannot price fall back to the custom-quote error by design.
2. Reductions never go below devices in use — a client must free seats first; the error message says exactly that.

3. The .lic endpoint never accepts a client-supplied fingerprint; misuse-by-migration stays impossible without the admin Shift-machine action.
 4. Renewal orders created *before* this deploy carry no `apply_device_limit` meta — they renew at the size they were billed at, exactly as before; schedules only affect renewal orders raised after deploy.
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Prepared by Claude (Cowork) for Ametecs Pvt Ltd — internal engineering report. SmartEPT Central remains the single commercial source of truth.